

## Basic Generic Definitions

### Example 1 : Generic Identity Function

The identityGB function works for any type:

|       |                                                  |
|-------|--------------------------------------------------|
| $[X]$ | $\text{identityGB} : X \rightarrow X$            |
|       | $\forall x : X \bullet \text{identityGB}(x) = x$ |

This defines identityGB polymorphically. It can be used with any type: identityGB applied to a natural number returns that number, identityGB applied to a set returns that set, etc.

### Example 2 : Generic Constant

A generic empty set:

|       |                               |
|-------|-------------------------------|
| $[X]$ | $\text{empty} : \mathbb{P} X$ |
|       | $\text{empty} = \{\}$         |

The empty set can be instantiated for any type X.

### Example 3 : Generic Pair Functions

Extract the first element of a pair:

|          |                                                     |
|----------|-----------------------------------------------------|
| $[X, Y]$ | $\text{fst} : X \times Y \rightarrow X$             |
|          | $\forall x : X; y : Y \bullet \text{fst}(x, y) = x$ |

Extract the second element:

|          |                                                     |
|----------|-----------------------------------------------------|
| $[X, Y]$ | $\text{snd} : X \times Y \rightarrow Y$             |
|          | $\forall x : X; y : Y \bullet \text{snd}(x, y) = y$ |

These work for pairs of any types.

### Example 4 : Generic Singleton Set

Create a set containing a single element:

|       |                                                     |
|-------|-----------------------------------------------------|
| $[X]$ | $\text{singleton} : X \rightarrow \mathbb{P} X$     |
|       | $\forall x : X \bullet \text{singleton}(x) = \{x\}$ |

Given any value, singleton produces a set containing just that value.

## Example 5 : Generic Sequence Operations

The head of a sequence (generic version):

|       |                                                                          |
|-------|--------------------------------------------------------------------------|
| $[X]$ | $\text{headOf} : \text{seq } X \rightarrow X$                            |
|       | $\forall s : \text{seq } X \mid \#s > 0 \bullet \text{headOf}(s) = s(1)$ |

This is defined only for non-empty sequences.

## Example 6 : Generic Set Membership ( Conceptual )

Set membership can be tested using the built-in 'in' operator in Z notation. In other contexts, you might define a membership test function, but Z notation provides this as a primitive predicate:  $x \text{ in } S$  is true exactly when  $x$  is an element of  $S$ .

## Example 7 : Generic Optional Type ( Conceptual )

Optional values can be modeled with free types in Z notation. For a specific type, you would define:  $\text{Option} ::= \text{none} \mid \text{some}(T)$ . Generic optional types require more complex setup beyond basic gendef blocks, as free types in Z are typically defined for specific types rather than being polymorphic.

## Example 8 : Generic Swap Function

Swap the components of a pair:

|          |                                                           |
|----------|-----------------------------------------------------------|
| $[X, Y]$ | $\text{swap} : X \times Y \rightarrow Y \times X$         |
|          | $\forall x : X; y : Y \bullet \text{swap}(x, y) = (y, x)$ |

Converts  $(x, y)$  to  $(y, x)$ .

## Example 9 : Generic Cardinality

Size of a set (for finite sets):

|       |                                                         |
|-------|---------------------------------------------------------|
| $[X]$ | $\text{size} : \mathbb{P} X \rightarrow \mathbb{N}$     |
|       | $\forall S : \mathbb{P} X \bullet \text{size}(S) = \#S$ |

Returns the number of elements in a finite set.

## Example 10 : Generic List Construction

Construct a singleton sequence:

|       |                                                              |
|-------|--------------------------------------------------------------|
| $[X]$ | $\text{single} : X \rightarrow \text{seq } X$                |
|       | $\forall x : X \bullet \text{single}(x) = \langle x \rangle$ |

Wraps a value in a single-element sequence.

### Example 11 : Using Generic Definitions

Once defined, generic functions can be instantiated for specific types:

$$\begin{array}{|l} id\_nat : \mathbb{N} \longrightarrow \mathbb{N} \\ id\_set : \mathbb{P} \mathbb{N} \longrightarrow \mathbb{P} \mathbb{N} \\ \hline id\_nat = identityGB[\mathbb{N}] \\ id\_set = identityGB[\mathbb{P} \mathbb{N}] \end{array}$$

Here  $identityGB[\mathbb{N}]$  is the  $identityGB$  function instantiated for natural numbers, and  $identityGB[\mathbb{P} \mathbb{N}]$  is instantiated for sets of natural numbers.

### Example 12 : Generic Constants

Define a generic empty sequence:

$$\begin{array}{|l} [X] \\ \hline emptySeq : seq \ X \\ \hline emptySeq = \langle \rangle \end{array}$$

This can be instantiated as  $emptySeq[\mathbb{N}]$  for an empty sequence of naturals,  $emptySeq[Person]$  for an empty sequence of persons, etc.